

Recitation: Rational Functions & Polynomial Long Division

Problem 1. Consider the rational function given below:

$$f(x) = \frac{3x^2 - 2x - 5}{(x + 2)(x^8 - 1)}$$

Complete the following parts:

- (a) Find the domain.
- (b) Find any y -intercepts.
- (c) Find any x -intercepts.
- (d) Find any vertical asymptotes.
- (e) Find any horizontal asymptotes.
- (f) Find any holes.
- (g) Can this rational function be polynomial long divided? Explain.

Problem 2. Complete the parts given below:

- (a) Factor the polynomial $x^3 - 25x$. Without explicitly computing, what is the remainder when this polynomial is divided by $x + 5$? Explain.
- (b) If $(3x^2 - x + 6)(x^2 + x - 3) + 2x - 4 = 3x^4 + 2x^3 - 4x^2 + 11x - 22$, then what is the polynomial quotient and remainder when $3x^4 + 2x^3 - 4x^2 + 11x - 22$ is divided by $(x^2 + x - 3)$? What about if it were instead divided by $3x^2 - x + 6$?
- (c) Without explicitly computing, can $x^2 - 4x + 6$ be divisible by $x^3 + 1$? Explain.
- (d) If $f(x)$ is divisible by $x - c$, what is $f(c)$? Explain. Is the 'reverse' of your conclusion true? Explain.

Problem 3. Polynomial long divide the expression below:

$$\frac{5x^4 - 15x^3 - x^2 + 12x - 27}{x - 3}$$

Problem 4. Polynomial long divide the expression below:

$$\frac{2x^5 + 3x^4 - 6x^3 - 10x^2 - x + 9}{x^2 - 3}$$

Problem 5. Use the fact that $x = 1$ is a root of $9x^3 - 3x^2 - 5x - 1$ to fully factor this polynomial.